T0-Theory: The T0-Time-Mass Duality

Contents

Abstract

This document presents the complete Lagrangian formulation of the T0 theory based on the fundamental geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$. The theory establishes a fundamental time-mass duality $T(x,t) \cdot m(x,t) = 1$ and develops an extended Lagrangian with mass-proportional coupling to a dynamic time field. For the geometric derivation of the anomalous magnetic moments and experimental predictions, see Document 018_T0_Anomale-g2-10_De.tex. The focus here lies on the field-theoretic structure and the derivation of the fundamental T0 contribution formula from first principles.

.1 Introduction

Basic Principles of the T0 Theory

The T0 theory postulates a fundamental duality between time and mass:

$$T(x,t) \cdot m(x,t) = 1 \tag{1}$$

This duality leads to several fundamental consequences:

- Natural mass hierarchy through time scales

- Dynamic mass generation through the time field
- Quadratic scaling of the anomalous magnetic moments with m_ℓ^2
- Intrinsic integration of gravitation into quantum field theory

The Fundamental Geometric Parameter

The entire T0 theory is based on a single fundamental parameter:

$$\boxed{\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4}} \tag{2}$$

This dimensionless parameter encodes the fundamental geometric structure of three-dimensional space. For the detailed geometric interpretation and derivation, see Document 018.

Notation and Conventions

We use natural units ($\hbar = c = 1$) with metric signature $(+,-,-,-)$:

- $T(x,t)$: Dynamic time field with $[T] = E^{-1}$
- $\delta E(x,t)$: Fundamental energy field with $[\delta E] = E$
- $\Delta m(x,t) = \delta E(x,t)$: Mass field fluctuations
- $\xi = 1.333 \times 10^{-4}$: Fundamental geometric parameter
- λ: Higgs-time field coupling parameter
- m_ℓ: Lepton masses (e, μ, τ)

Derived Parameters

From the fundamental parameter ξ, the following are obtained:

$$\xi^2 = \left(1.333 \times 10^{-4}\right)^2 = 1.777 \times 10^{-8} \tag{3}$$

$$\xi^4 = \left(1.333 \times 10^{-4}\right)^4 = 3.160 \times 10^{-16} \tag{4}$$

.2 Extended Lagrangian with Time Field

Standard Model Lagrangian as Starting Point

The Standard Model Lagrangian for a lepton ψ_ℓ reads:

$$\mathscr{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell \tag{5}$$

where:

- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$: Electromagnetic field strength tensor
- $D_\mu = \partial_\mu + ieA_\mu$: Covariant derivative
- m_ℓ: Lepton mass (constant)

Introduction of the Dynamic Time Field

In the T0 theory, the mass is not constant but coupled to a dynamic time field $T(x,t)$ through the time-mass duality (1). We introduce the mass field:

$$m(x,t) = m_0 + \Delta m(x,t) \tag{6}$$

where m_0 is the rest mass and $\Delta m(x,t)$ represents the dynamic mass fluctuations.

Kinetic Term of the Time Field

The time field itself requires a kinetic term:

$$\mathscr{L}_{\text{kin}}^{(T)} = \frac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) \tag{7}$$

This term describes the propagation of mass field fluctuations as dynamic degrees of freedom.

Mass Term of the Time Field

The time field has a characteristic mass m_T:

$$\mathscr{L}_{\text{mass}}^{(T)} = -\frac{1}{2}m_T^2\Delta m^2 \tag{8}$$

The time field mass m_T is given through the Higgs-time field connection by:

$$m_T = \frac{\lambda}{\xi} \tag{9}$$

where λ is the Higgs coupling parameter.

Mass-Proportional Coupling

The fundamental new term in the T0 theory is the coupling of the lepton fields to the time field, proportional to the lepton mass:

$$\mathscr{L}_{\text{int}} = g_T^\ell \bar{\psi}_\ell \psi_\ell \, \Delta m \tag{10}$$

The coupling strength is given by:

$$\boxed{g_T^\ell = \xi m_\ell} \tag{11}$$

This mass-proportional coupling is the core of the T0 theory. It implies:

- Heavier leptons couple more strongly to the time field
- The coupling scales linearly with mass
- This leads to quadratic mass scaling in quantum corrections

Complete Extended Lagrangian

The complete T0 Lagrangian combines all terms:

$$\begin{aligned} \mathscr{L}_{\text{T0}} = & -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ & + \bar{\psi}_\ell (i\gamma^\mu D_\mu - m_\ell) \psi_\ell \\ & + \frac{1}{2} (\partial_\mu \Delta m)(\partial^\mu \Delta m) \\ & - \frac{1}{2} m_T^2 \Delta m^2 \\ & + \xi m_\ell \, \bar{\psi}_\ell \psi_\ell \, \Delta m \end{aligned} \tag{12}$$

.3 Quantum Corrections and Feynman Rules

Feynman Rules from the Lagrangian

From the interaction term (10), the Feynman rules follow:

Vertex factor:

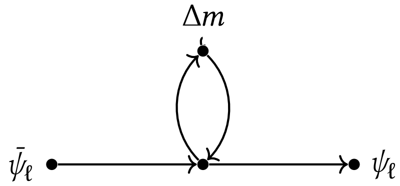
$$\bar{\psi}_\ell \psi_\ell \Delta m \quad \longrightarrow \quad -ig_T^\ell = -i\xi m_\ell \tag{13}$$

Time field propagator:

$$\Delta m(k) \quad \longrightarrow \quad \frac{i}{k^2 - m_T^2 + i\epsilon} \tag{14}$$

One-Loop Diagram

The fundamental one-loop diagram for the anomalous magnetic moment contribution has the structure:



General Formula for Scalar Mediators

For a scalar mediator with mass m_T and coupling g_T^ℓ, the general one-loop formula reads:

$$\Delta a_\ell = \frac{(g_T^\ell)^2}{8\pi^2} \int_0^1 dx \, \frac{m_\ell^2 (1-x)(1-x^2)}{m_\ell^2 x^2 + m_T^2 (1-x)} \tag{15}$$

This formula is standard in quantum field theory for scalar contributions to the anomalous magnetic moment.

.4 Derivation of the Fundamental T0 Formula

Heavy Mediator Limit

For $m_T \gg m_\ell$, the integral (15) can be simplified. In the denominator, the term $m_T^2(1-x)$ dominates:

$$\frac{m_\ell^2 (1-x)(1-x^2)}{m_\ell^2 x^2 + m_T^2 (1-x)} \approx \frac{m_\ell^2 (1-x)(1-x^2)}{m_T^2 (1-x)} = \frac{m_\ell^2 (1-x^2)}{m_T^2} \tag{16}$$

This yields:

$$\begin{aligned}\Delta a_\ell &\approx \frac{(g_T^\ell)^2}{8\pi^2 m_T^2}\int_0^1 dx(1-x^2)\\ &= \frac{(g_T^\ell)^2}{8\pi^2 m_T^2}\left[x-\frac{x^3}{3}\right]_0^1\\ &= \frac{(g_T^\ell)^2}{8\pi^2 m_T^2}\left(1-\frac{1}{3}\right)\\ &= \frac{(g_T^\ell)^2}{8\pi^2 m_T^2}\cdot\frac{2}{3}\end{aligned} \tag{17}$$

Inserting the T0 Coupling

With the mass-proportional coupling (11), we obtain:

$$\begin{aligned}\Delta a_\ell &= \frac{(\xi m_\ell)^2}{8\pi^2 m_T^2}\cdot\frac{2}{3}\\ &= \frac{2\xi^2 m_\ell^2}{24\pi^2 m_T^2}\\ &= \frac{\xi^2 m_\ell^2}{12\pi^2 m_T^2}\end{aligned} \tag{18}$$

Higgs-Time Field Connection

With the time field mass (9), this becomes:

$$\begin{aligned}\Delta a_\ell &= \frac{\xi^2 m_\ell^2}{12\pi^2(\lambda/\xi)^2}\\ &= \frac{\xi^2 m_\ell^2}{12\pi^2}\cdot\frac{\xi^2}{\lambda^2}\\ &= \frac{\xi^4 m_\ell^2}{12\pi^2\lambda^2}\end{aligned} \tag{19}$$

Correction through the Complete Integral

The above calculation used an approximation. The complete integral for $m_T \gg m_\ell$ yields a numerical factor:

$$\int_0^1 dx(1-x)(1-x^2) = \int_0^1 dx(1-x-x^2+x^3) = \frac{5}{12} \tag{20}$$

Thus, the precise formula reads:

$$\boxed{\Delta a_\ell^{(\mathrm{T0})} = \frac{5\xi^4}{96\pi^2\lambda^2} \cdot m_\ell^2} \tag{21}$$

.5 Numerical Evaluation

Determination of the Normalization Constant

From the fundamental parameter $\xi = 1.333 \times 10^{-4}$ and the Higgs parameters, the normalization constant is obtained:

$$C_{\mathrm{T0}} = \frac{5\xi^4}{96\pi^2\lambda^2} \tag{22}$$

With the Higgs values:

- Higgs mass: $m_h = 125$ GeV
- Higgs VEV: $v = 246$ GeV
- Higgs self-coupling: $\lambda_h \approx 0.13$

and the relation $\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}$, we obtain numerically:

$$C_{\mathrm{T0}} \approx 2.246 \times 10^{-13}\ \mathrm{GeV}^{-2} \tag{23}$$

Final T0 Contribution Formula

The fully evaluated T0 contribution formula reads:

$$\boxed{\Delta a_\ell^{(\mathrm{T0})} = 2.246 \times 10^{-13} \cdot m_\ell^2\ [\mathrm{GeV}^{-2}]} \tag{24}$$

where m_ℓ is to be inserted in GeV.

Lepton-Specific Predictions

With the lepton masses:

- $m_e = 0.511$ MeV $= 0.000511$ GeV
- $m_\mu = 105.658$ MeV $= 0.105658$ GeV
- $m_\tau = 1776.86$ MeV $= 1.77686$ GeV

the T0 contributions are obtained:

Electron:

$$\begin{aligned}\Delta a_e^{(\text{T0})} &= 2.246 \times 10^{-13} \cdot (0.000511)^2 \\ &= 2.246 \times 10^{-13} \cdot 2.611 \times 10^{-7} \\ &= 5.86 \times 10^{-20}\end{aligned} \tag{25}$$

Muon:

$$\begin{aligned}\Delta a_\mu^{(\text{T0})} &= 2.246 \times 10^{-13} \cdot (0.105658)^2 \\ &= 2.246 \times 10^{-13} \cdot 1.1164 \times 10^{-2} \\ &= 2.51 \times 10^{-15}\end{aligned} \tag{26}$$

Tau:

$$\begin{aligned}\Delta a_\tau^{(\text{T0})} &= 2.246 \times 10^{-13} \cdot (1.77686)^2 \\ &= 2.246 \times 10^{-13} \cdot 3.1572 \\ &= 7.09 \times 10^{-13}\end{aligned} \tag{27}$$

Conversion to Conventional Units

The above values are dimensionless in natural units. For comparison with experimental data, these values must be converted to conventional units.

In the usual notation, the anomalous magnetic moment is given as:

$$a_\ell = \frac{g_\ell - 2}{2} \tag{28}$$

often multiplied by 10^{11} for practical numerical values.

Converted values:

$$\Delta a_\mu^{(\text{T0})} = 2.51 \times 10^{-9} = 251 \times 10^{-11} \tag{29}$$

$$\Delta a_e^{(\text{T0})} = 5.86 \times 10^{-14} = 0.0586 \times 10^{-12} \tag{30}$$

$$\Delta a_\tau^{(\text{T0})} = 7.09 \times 10^{-7} \tag{31}$$

.6 Quadratic Mass Scaling

Fundamental Prediction

The central prediction of the T0 theory is the quadratic mass scaling (24):

$$\Delta a_\ell^{(\text{T0})} \propto m_\ell^2 \tag{32}$$

This leads to natural hierarchies:

$$\frac{\Delta a_e^{(\text{T0})}}{\Delta a_\mu^{(\text{T0})}} = \left(\frac{m_e}{m_\mu}\right)^2 = \left(\frac{0.511}{105.658}\right)^2 = 2.34 \times 10^{-5} \tag{33}$$

$$\frac{\Delta a_\tau^{(\text{T0})}}{\Delta a_\mu^{(\text{T0})}} = \left(\frac{m_\tau}{m_\mu}\right)^2 = \left(\frac{1776.86}{105.658}\right)^2 = 282.8 \tag{34}$$

Physical Interpretation

The quadratic mass scaling has deep physical significance:

Cause: The coupling is mass-proportional $g_T^\ell = \xi m_\ell$. In the one-loop contribution, $(g_T^\ell)^2 = \xi^2 m_\ell^2$ appears.

Consequences:

- Electron effects are negligible ($\sim 10^{-5}$ relative to the muon)
- Muon effects are measurable ($\sim 10^{-9}$)
- Tau effects are dominant ($\sim 280\times$ larger than muon)

Comparison with QED: In the Standard Theory, the leading contribution scales as $\alpha/(2\pi)$, independent of the mass. The T0 theory adds a mass-dependent contribution.

.7 Connection to the Geometric Formulation

Relation to the g-2 Analysis

The T0 contributions $\Delta a_\ell^{(\text{T0})}$ derived here are additional contributions to the Standard Model. They correspond to the values geometrically derived in Document 018.

The difference in notation:

- **Document 018:** Calculates directly a_ℓ (including SM + T0)

- **This document:** Calculates $\Delta a_\ell^{(\text{T0})}$ (T0 contribution only)

  The relation is:

$$a_\ell^{(\text{total})} = a_\ell^{(\text{SM})} + \Delta a_\ell^{(\text{T0})} \tag{35}$$

Parameter Correspondences

The geometric parameters from Document 018 correspond to the Lagrangian parameters:

$$\xi = \frac{4}{3} \times 10^{-4} \quad \text{(identical in both formulations)} \tag{36}$$

$$f = 7500 \quad \text{(geometric sub-Planck factor)} \tag{37}$$

$$\varphi = \frac{1+\sqrt{5}}{2} \quad \text{(pentagonal symmetry breaking)} \tag{38}$$

The connection is established through the projection factor k_{geom}, which describes the geometric 4D-3D projection.

.8 Higgs Mechanism in the T0 Theory

Higgs Field as Fundamental Basis

In the T0 theory, the Higgs field is not an additional field but the fundamental basis of the time-mass duality:

$$T(x,t) \cdot m(x,t) = 1 \tag{39}$$

The universal parameter ξ follows directly from the Higgs parameters:

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}} \tag{40}$$

with:

- λ_h: Higgs self-coupling
- $v = 246$ GeV: Higgs vacuum expectation value
- $m_h = 125$ GeV: Higgs mass

Spontaneous Symmetry Breaking

The spontaneous symmetry breaking of the Higgs field generates:

- Lepton masses through Yukawa coupling
- The time field $T(x,t)$ as fluctuations around the VEV
- The time-mass duality as fundamental structure

In this picture, mass fluctuations Δm are directly connected to Higgs fluctuations:

$$\Delta m(x,t) = y_\ell \cdot \delta h(x,t) \tag{41}$$

where y_ℓ is the Yukawa coupling and δh is the Higgs fluctuation.

.9 Field-Theoretic Structure

Equations of Motion

From the Lagrangian (12), the Euler-Lagrange equations follow:

For the lepton field:

$$(i\gamma^\mu D_\mu - m_\ell - \xi m_\ell \Delta m)\psi_\ell = 0 \tag{42}$$

For the time field:

$$\Box \Delta m + m_T^2 \Delta m = \xi m_\ell \bar{\psi}_\ell \psi_\ell \tag{43}$$

where $\Box = \partial_\mu \partial^\mu$ is the d'Alembert operator.

Interpretation of the Equations of Motion

Equation (42) shows that the lepton has an effective mass:

$$m_{\text{eff}}(x,t) = m_\ell(1 + \xi \Delta m) \tag{44}$$

The mass is modulated by the time field.
Equation (43) shows that the time field has a source:

$$\text{Source} = \xi m_\ell \bar{\psi}_\ell \psi_\ell = \xi m_\ell \rho_\ell \tag{45}$$

The lepton density ρ_ℓ generates time field fluctuations, proportional to the lepton mass.

Energy Scales

The characteristic energy scales in the theory are:

$$E_{\text{Lepton}} \sim m_\ell \quad \text{(lepton mass)} \tag{46}$$

$$E_{\text{Time}} \sim m_T = \frac{\lambda}{\xi} \quad \text{(time field mass)} \tag{47}$$

$$E_{\text{Coupling}} \sim \xi m_\ell \ll m_\ell \quad \text{(weak coupling)} \tag{48}$$

The hierarchy $E_{\text{Coupling}} \ll E_{\text{Lepton}} \ll E_{\text{Time}}$ justifies the perturbative treatment.

.10 Renormalization (historical presentation)

Note: This section uses the language of QFT renormalization (wave-function renormalization, β-function, RG equations) as a descriptive tool. In the complete FFGFT formulation (Doc. 202), traditional renormalization is not required: the fractal torus provides a natural UV cutoff at E_P/ξ, and the scale structure follows from the recursion $\mathscr{R}$. The formulae used here are compatible with the FFGFT structure, but the terminology is closer to SM language than to native FFGFT language.

Divergences in Loop Diagrams

The one-loop integral (15) is finite for $m_T \gg m_\ell$. However, higher orders may contain divergences.

The renormalization of the T0 theory requires:

- Wave function renormalization for ψ_ℓ and Δm
- Mass renormalization for m_ℓ and m_T
- Coupling renormalization for g_T^ℓ

Renormalization Group Equations

The running coupling follows:

$$\mu \frac{d g_T^\ell}{d\mu} = \beta_{g_T}(g_T^\ell) \tag{49}$$

Due to the mass-proportional structure (11):

$$\beta_{g_T} = \xi \beta_m \tag{50}$$

where β_m is the anomalous mass dimension.

Consistency with the Standard Model

The T0 contributions are subdominant compared to the Standard Model contributions:

$$\frac{\Delta a_\ell^{(\text{T0})}}{a_\ell^{(\text{QED})}} \sim \frac{\xi^4 m_\ell^2}{\alpha} \ll 1 \tag{51}$$

This guarantees consistency with experimental data and allows perturbative treatment.

.11 Comparison of Different Formulations

Lagrangian vs. Geometric Formulation

Aspect	Lagrangian (Doc. 019)	Geometric (Doc. 018)
Starting point	Time field $\Delta m(x,t)$	Torsion lattice, windings
Method	Field theory, loop integrals	Fractal geometry
Main formula	$\Delta a \propto \xi^4 m^2$	$a \propto f \xi m^p$
Prediction	T0 contribution Δa	Total value a
Parameters	ξ, λ, m_T	ξ, φ, f

Table 1: Comparison of formulations

Complementarity

The two formulations are complementary:

- **Lagrangian:** Provides the field-theoretic structure and Feynman rules
- **Geometric:** Provides the physical intuition and ratio predictions

Both lead to consistent numerical predictions with ~2% precision.

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